

Name: _____



Expressing multiplicative relationships as ratios and fractions

Task A: Multiplicative relationships as ratios

1) Using the ratio table, fill in the blanks.

Flour (g)	Eggs	Water (ml)	Sugar (g)
100	1	25	200

a) Number of eggs $\times$ $\frac{\quad}{\quad}$ = Flour (g)

For every egg, there is $\frac{\quad}{\quad}$ g of flour

b) Amount of water (ml) $\times$ $\frac{\quad}{\quad}$ = Sugar (g)

For every ml of water, there is $\frac{\quad}{\quad}$ of flour

c) Amount of sugar (g) $\times$ $\frac{\quad}{\quad}$ = Flour (g)

For every 1 g of sugar, $\frac{\quad}{\quad}$

2) Using the following bar models, fill in the gaps.

S	S	S	S	S	S	T
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a) Shorts $\times$ = T-shirts

For every pair of shorts, there are

C	C	M	M	M	M
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b) Cat $\times$ = Mice

For every cat, there are

H	H	P	P	P	P
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c) House $\times$ = Phone

For every house , there are



Task A: Multiplicative relationships as ratios

3) Jacob spills ink all over his work.

Can you work out what was under the ink splats?

Here is a bar model showing the ratio water, glue and paper.



Ratio table

W	G	P
2		

For every 1 part water, there are 3 parts glue.

For every 1 part paper, there are 2 parts glue.

For every 1 part glue, there are 0.5 parts .

For every there are 1.5 parts paper.

4) Laura, Lucas and Alex are making papier mache.

The instructions are: for every 4 tablespoons of white flour use 240 ml of water.

Laura used 8 tablespoons of flour and 480 ml of water.

Alex used 6 tablespoons of flour and 242 ml of water.

Lucas used 2 tablespoons of flour and 238 ml of water.

a) Who used the correct ratio for making papier mache? Explain the error of the other pupils.

b) Whose mixture will be too watery? Explain why.



Task B: Multiplicative relationships as fractions

1) Using the bar model, fill in the gaps.

a)

C	D	D	D
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 Fraction of dogs =

Cat $\times$ = DogFor every cat, there are dogs

b)

P	P	H	H	H	H	H
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 Fraction of hearts =

Present $\times$

 = HeartFor every present, there are hearts

c)

A	O	A	O	O	O
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 Fraction of oranges =

Apple $\times$ = OrangeFor every apple, there are oranges

2) There are four types of smoothies for sale all containing strawberries, apples and banana.

A
For every strawberry, there are 2 slices of apples and 5 slices of banana.

B
All fruit is equally split.

C
Banana $\times \frac{1}{3}$ = Apples
For every strawberry, there are 2 slices of banana.

D
Apple and banana have the same ratio.
Strawberry is 50% of banana.

a) Jacob likes strawberries, which smoothie should he pick?

b) Sofia likes apple, which smoothie should she pick?

c) Which smoothie has the least banana?

d) Create your own smoothie so the ratio of banana > ratio of apple > ratio of strawberries.